# Anthracite

△ **Specimen** of anthracite showing its semi-metallic lustre

**Anthracite is the purest**, most carbonized type of coal, consisting almost entirely of carbon. Like bituminous coal, it is composed of organic matter, but is older and much more highly compressed, and does not leave behind any residue when touched. It is used in beads and carvings, although its main use is as a fuel – though difficult to ignite, once lit it produces a lot of heat and burns slowly. Anthracite fires combust with a small blue flame that is smokeless, making it a good fuel for indoor use, but its expense means it is less widely used on an industrial level.

## Specification

**Chemical name** Anthracite | **Formula** $C_{240}H_{90}O_4NS$ **Colours** Metallic black | **Structure** Amorphous | **Hardness** 2.75–3 | **SG** 1.4 | **RI** 1.64–1.68 | **Lustre** Sub-metallic **Streak** n/a | **Locations** Russia, Ukraine, North Korea, South Africa, Vietnam, UK, Australia, USA


**Compact form** | **Rough** | The typical density of anthracite can be seen in this specimen, as can its characteristic, near-metallic, lustre.


**Bright lustre** | **Rough** | The irregular surface of this specimen of anthracite shows an unusually bright lustre and inclusions of rock groundmass.

**Blocky breakage** | **Rough** | Because anthracite is hard and brittle, its surface tends to break in sharply angled blocks, as in this specimen.


**Contrasting specimens** | **Rough** | The upper specimen is bituminous, or ordinary household, coal, while the lower specimen is anthracite.

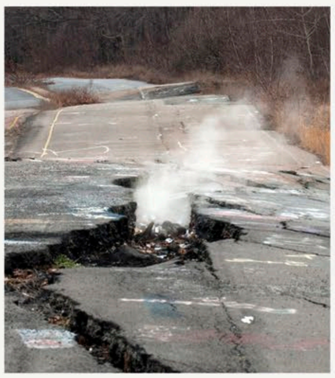
**Weathered anthracite** | **Rough** | When exposed to weathering, the outer layers of anthracite blocks oxidize and deteriorate, as in this specimen.


**Polished anthracite** | **Cut** | This irregularly shaped piece of anthracite is polished to a sheen, showing how it can sometimes be used as a jet-substitute.

## Slow burn

### The Centralia mine fire

An underground fire has been burning for decades in an anthracite mine in Centralia, Pennsylvania, USA. The fire started in 1962, and came to a head in 1981 when a 12-year-old fell into a 46m (50yd) sinkhole caused by the fire, which opened up beneath him (he survived, hauled out by his cousin with a rope). The fire is still burning and Centralia is now a ghost town.



**The Centralia fire** Anthracite burning in the old mine can be seen here breaking through the surface of the ground.